Computer Organization
Homework 4
(Due on 2018/12/20)

Note: Please hand in the hardcopy before the due day.

1. IEEE 754-2008 contains a half precision that is only 16 bits wide. The leftmost bit is still the sign bit, the exponent is 5 bits wide and has a bias of 15, and the fraction is 10 bits long. A hidden 1 is assumed. What is maximum and minimum of this representation for the positive values? (10%)

2. Calculate $(4.12109375 \times 10^{-1} + 3.45703125 \times 10^{-1}) + 2.143 \times 10^{3}$ by hand, assuming all the values are stored in the 16-bit half precision format described in Q1. Assume 1 guard bit, 1 round bit, and 1 sticky bit, and round to the nearest even. Show all the steps, and write your answer in both the 16-bit floating point format and in decimal. (10%)

3. Calculate $4.12109375 \times 10^{-1} + (3.45703125 \times 10^{-1} + 2.143 \times 10^{3})$ by hand, assuming all the values are stored in the 16-bit half precision format described in Q1. Assume 1 guard bit, 1 round bit, and 1 sticky bit, and round to the nearest even. Show all the steps, and write your answer in both the 16-bit floating point format and in decimal. (10%)

4. The basic single-cycle MIPS implementation in Fig 4.2 (page 6 of the slides) can only implement some instructions. New instructions can be added to an existing instruction set architecture (ISA), but the decision whether or not to do that depends on the cost and complexity that are introduced into the processor datapath and control. Given the new instruction:
   Instruction: LWI Rt, Rd(Rs)
   Interpretation: Reg[Rt]= Mem[Reg[Rs]+Reg[Rd]]
   (a) Which existing blocks (if any) can be used for this instruction? (5%)
   (b) Which new functional blocks (if any) do we need for this instruction? (5%)
   (c) What new signals do we need (if any) from the control unit to support this instruction? (5%)

5. Consider the following instruction:
   Instruction: addi Rt, Rs, 40
   Interpretation: Reg[Rt]=Reg[Rs] +40

(a) What are the values of control signals generated by the control on page 24 of the slides) for the above instruction? (7 control signals) (15%)
(b) Which blocks perform a useful arithmetic function for this instruction? (5%)
(c) Which blocks produce outputs, but their outputs are not used for this instruction? (5%) Which blocks produce no outputs for this instruction? (5%)

6. Problems in this exercise assume that logic blocks needed to implement a processor's datapath have the following latencies:

I-Mem Read/ Write	Add	Mux	ALU	Regs Read/ Write	D-Mem Read/ Write	Sign-Extend	Sift-left-2
240ps	50ps	40ps	120ps	120ps	280ps	30ps	20ps

(a) If the only thing we need to do in a processor is fetch consecutive instruction, what would the cycle time be? (Page 13 of the slides) (5%)
(b) Consider a datapath similar to the one in Fig. 4.11 (Page 20 of the slides), but for a processor that only has R-ype of instruction. What would the cycle time be for this datapath? (5%)
(c) For the datapath element Shift-left-2, which kinds of instructions require this resource block? (5%) What would the cycle time be for this kind of instructions? (5%)
(d) Consider a datapath similar to the one in Fig. 4.11 (Page 20 of the slides), but for a processor that needs to support R-type, load/save, branch instructions. What would the cycle time be for this datapath? (5%)
(e) From (d), for which kinds of instructions (if any) is this resource on the critical path. (5%)